Unit 2, Lesson 8: Dividing Positive Fractions

Benchmark

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.

Learning Target

- ☐ I can fluently divide positive fractions by positive fractions.

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2.8.1 Warm-Up: Error Analysis

Mateo tried to multiply $3\frac{5}{6} \cdot 4\frac{1}{10}$, but he made an error. His work is shown.

$$3\frac{5}{6} \cdot 4\frac{1}{10} = 12\frac{5}{60} = 12\frac{1}{12}$$

1. What error did Mateo make?

2. What is something that Mateo did correctly?

3. Help Mateo understand and learn from his mistake. Write a brief note to Mateo explaining the mistake he made and how to fix it moving forward.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

4. Find the product of $3\frac{5}{6} \cdot 4\frac{1}{10}$. Show your work.

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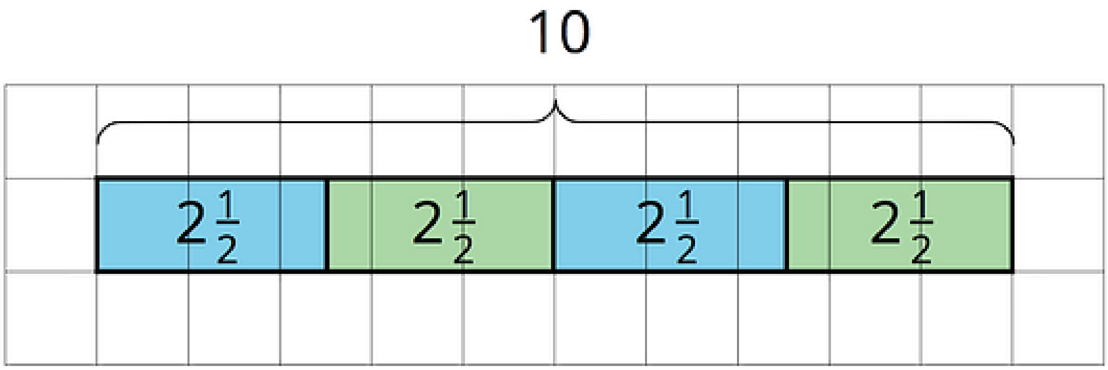
2.8.2 Guided Instruction: Dividing Fractions

Recall how tape diagrams can be used to illustrate division.



1. Write the division equation represented by the tape diagram.

dividend		divisor		quotient
________	÷	________	=	________

2. How many groups of $2\frac{1}{2}$ are in 10?

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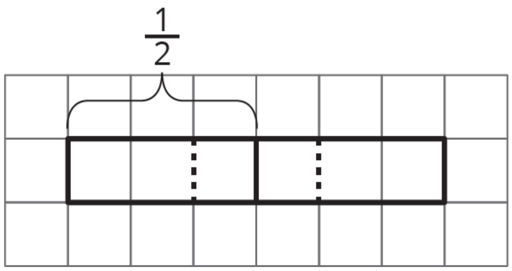
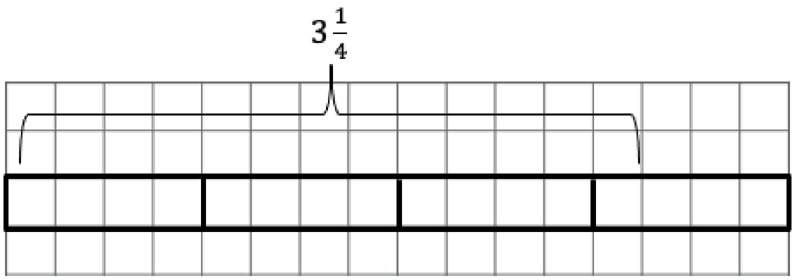
3. Complete the table by finding each quotient. Use the tape diagram shown.

Tape Diagram	Division Expression	Quotient
	$\frac{1}{2} \div \frac{1}{3}$	
	$3\frac{1}{4} \div \frac{3}{4}$	

4. Jacquelyn is making ground beef flautas, where each flauta will use $\frac{3}{10}$ pounds of ground beef. She has a total of $1\frac{4}{5}$ pounds of ground beef to use.

a. Which division statement below represents the number of flautas Jacquelyn can make?

$\frac{3}{10} \div 1\frac{4}{5}$ $\qquad$ $1\frac{4}{5} \div \frac{3}{10}$

b. Draw a tape diagram to illustrate the division statement you chose.

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c. Solve the division statement.

d. How many $\frac{3}{10}$ pound flautas can she make?

You previously developed an algorithm to use when dividing fractions.

When **dividing fractions**, the division statement can be rewritten as an equivalent multiplication sentence by using the **reciprocal**, or multiplicative inverse, of the second fraction.

$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$

Existence of multiplicative inverses

For every $a \neq 0$ there exists $\frac{1}{a}$ so that $a \times \frac{1}{a} = 1$ and $\frac{1}{a} \times a = 1$.

With the tape diagrams, you determined $3\frac{1}{4} \div \frac{3}{4} = 4\frac{1}{3}$.

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The standard algorithm can also be used to determine the quotient.

Write the given division expression.	$3\frac{1}{4} \div \frac{3}{4}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{13}{4} \div \frac{3}{4}$
Rewrite the division using multiplication by the reciprocal of the second fraction.	$\frac{13}{4} \times \frac{4}{3}$
Multiply the fractions and simplify.	$\frac{52}{12} = \frac{13}{3} = 4\frac{1}{3}$

5. Find the quotient $5\frac{2}{3} \div 2\frac{4}{9}$ using the algorithm.

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2.8.3 Guided Practice: Dividing Fractions

1. Complete the table to find each quotient.

Division Sentence	Rewritten as Fractions (If Needed)	Equivalent Multiplication Sentence	Quotient
$\frac{7}{8} \div 2\frac{3}{10}$			
$\frac{2}{7} \div \frac{3}{4}$			
$8\frac{2}{3} \div \frac{5}{6}$			
$4\frac{4}{5} \div 1\frac{1}{2}$			

2. Find each quotient using your preferred method. Simplify when possible.

a. $\frac{3}{4} \div \frac{9}{14}$

b. $6\frac{7}{8} \div \frac{5}{7}$

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2.8.4 Your Turn!

1. Find each quotient. Simplify when possible.

a. $\frac{1}{18} \div \frac{1}{9}$

b. $2\frac{2}{5} \div \frac{1}{10}$

c. $\frac{3}{15} \div 6$

d. $9 \div \frac{3}{2}$

e. $1\frac{5}{9} \div 5\frac{5}{8}$

f. $\frac{1}{20} \div \frac{1}{4}$

g. $\frac{4}{12} \div \frac{1}{7}$

h. $2 \div 3\frac{4}{15}$

i. $1\frac{6}{10} \div 5\frac{3}{4}$

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2.8.5 Wrap-Up: Ticket Out the Door

1. Find each product or quotient. Simplify when possible.

   a. $\frac{2}{11} \div \frac{1}{3}$

   b. $\left(\frac{2}{23}\right)\left(3\frac{3}{23}\right)$

   c. $2\frac{27}{32} \times \frac{3}{182}$

   d. $3\frac{3}{4} \div 4\frac{1}{6}$

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